

# Web Development

## HTML

It stands for Hypertext Markup Language

- Structural language
- Holds content
- Organizes items relative to each other
  - Think of this like putting boxes in a box to separate things inside
  - Or like a house, defines the rooms/walls

## Tags

`<tag></tag>` : tag pair

- This is a basic HTML tag
- Holds content for you or other tags

`<tag/>` : self closing tag

- Special tags close themselves
- This usually displays content

`<tag property="value" />` : tag with properties

- Some tags require more information
- Properties define deeper attributes
  - e.g. Class -> to help define styling in CSS
  - e.g. Src -> Images need a source file to show
  - e.g. Href -> Anchor tags (hyperlinks) need a destination to take you

## Basic Structure

```
<html>
  <head>
    This holds metadata, just more info about the page you can't see
    <link href='styles.css' rel='stylesheet' />
  </head>
  <body>
    This holds your content that you'd want to show
```

```
<p>paragraph/text</p>
<h1>heading/titles</h1>
<img src='img.png' /> shows an image
<ul> unordered list
  <li>list item</li>
</ul>
<div>used for organizing</div>
<a href='my_link.com'>hyperlink</a>
</body>
</html>
```

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# CSS

It stands for Cascading Style Sheet

- As the name suggests, it styles your HTML page
  - Think of it like a house, the wallpaper or paint
  - Sometimes tells you where to put the couch in the room
- Can move content in reference to other tags
- Manipulates fonts, colors, sizes, etc.

We use selectors to decide how specific we want to style our page

## \* selector

The star selector, or the style EVERYTHING selector

- This defines a global style that applies to everything on the page
- Would not recommend using this unless you know a consistent style is used

## tag selector

The tag selector

- Styles all tags of the same type
  - If you want all your paragraphs to have the same font
  - Or all your images a consistent size

## . selector

The class selector

- Group up HTML elements by class
  - `<tag class="some_class"/>`
- This allows all tags that have this class to style consistently

## # selector

The ID selector

- Unique to a tag
- Only if you want to change the style of something specific on your page

## Basic Structure

```
* {  
  color: blue;  
}  
  
h1 {  
  font-size: 32;  
}  
  
.class {  
  width: 10%;  
}  
  
#id {  
  display: flex;  
}
```